

Teodora M. Piel

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GitHub: <https://github.com/AuroraXP> Web: <https://tmpiel.com>

PROFILE

Career changer moving fully into software engineering, holding an M.Sc. from RWTH Aachen (completed May 2026) with a thesis that was pure software development: designing and building an ML-based automation tool in Python, with a working prototype plugin for established crystallography software. Self-taught programmer with proven ability to master new technical fields quickly and independently. Seeking a junior software engineering role; **available immediately** for remote positions and at short notice for on-site roles.

TECHNICAL AND LANGUAGE SKILLS

- **Programming:** Python (primary – automation, machine learning with scikit-learn, data analysis); TypeScript/JavaScript (basics)
- **Web:** Astro, HTML/CSS, Tailwind CSS, basic Solid.js
- **Tools & environment:** Unix-like environments (macOS, Linux/WSL), bash, Git, AI-assisted development (Claude), LaTeX
- **Languages (fluent):** German, English, Romanian

SOFTWARE & RESEARCH PROJECTS

Automation of Atom Type Assignment Process

Nov 2025 – May 2026

Master's Thesis, group of Prof. F. Kleemiß, RWTH Aachen

- Implemented SOAP-based statistical descriptors as input features and trained supervised machine-learning models on them (random forest and multi-layer perceptron classifiers; scikit-learn) to automate atom-type assignment in crystal structure refinement – a step most researchers still perform manually – improving on existing automated approaches
- Built validation logic that detects and corrects misassigned atoms in human-prepared structures, and suggests alternative assignments ranked by model confidence
- Developed a working prototype integration of the tool as a plugin for **Olex2**, an established crystallography software used worldwide (full integration and open-source release planned)

Benchmarking of a Novel Partitioning Scheme for Crystal Refinement

Apr 2025 – Aug 2025

Research Assistant, group of Prof. F. Kleemiß, RWTH Aachen

- Built and ran systematic benchmarks in Python to evaluate a new partitioning scheme against established methods
- Contributed to a peer-reviewed publication in *Z. Kristallogr. Cryst. Mater.* (see Publications)

Personal Website

Aug 2026 – ongoing

Personal project <https://tmpiel.com> – <https://github.com/AuroraXP/dora-web>

- Designed and built a personal portfolio site with Astro, Solid.js, and Tailwind CSS; deployed via Cloudflare Pages

EDUCATION

M.Sc. Chemistry, RWTH Aachen University

2024 – 2026

Thesis: Implementation of SOAP Statistical Descriptors for Improved Atom Type Assignment

B.Sc. Chemistry, RWTH Aachen University

2020 – 2024

Thesis: Synthesis and characterization of titanium(IV) complexes for PET scan applications

PUBLICATIONS

- Seifert, L. M., Brüx, D., **Piel, T. M.**, Kleemiß, F. "Grid-free quantum crystallographic refinement using unbiased analytic atomic form factors from density fitting." *Zeitschrift für Kristallographie – Crystalline Materials*, 2026. <https://doi.org/10.1515/zkri-2026-0013>

FURTHER RESEARCH EXPERIENCE

- **Research Assistant**, Inorganic Chemistry (Prof. I. Oppel), RWTH Aachen *Mar – Oct 2024*
Research and Synthesis of ligands and titanium(IV) complexes for PET scan applications
- **Research Internship**, Solid-State Chemistry (Prof. R. Dronskowski), RWTH Aachen *May – Sep 2024*
Synthesis and characterization of novel quaternary tellurides
- **Student Research Assistant**, Solid-State Chemistry (Prof. R. Dronskowski), RWTH Aachen *Apr – Jun 2023*
Synthesis of novel quaternary tellurides

INTERESTS AND HOBBIES

- Ice skating, painting, creative writing, baking, gaming